

What Is Middleware?

In ASP.NET Core, middleware are software components that process HTTP requests and responses.

Think of them as a pipeline of delegates, where each component:

1. Performs some operation on the incoming request.
2. Optionally passes it to the next middleware in the chain.
3. Performs some operation on the outgoing response.

Common examples include:

- Authentication & Authorization
- Logging
- Error Handling
- Routing
- Static Files

In short:

Middleware = Modular building blocks that define how requests are handled before reaching your application logic.

Where Does Kestrel Fit In?

Kestrel is not middleware — it's the **web server** that hosts your ASP.NET Core application and manages the low-level communication with the operating system.

When an HTTP request arrives:

1. **Kestrel** receives the request over TCP.
2. It **forwards** that request into the **ASP.NET Core middleware pipeline**.
3. Each middleware runs in order until one generates a response (or passes it onward).
4. The response then flows back through the pipeline and returns to Kestrel.
5. Kestrel sends the final response to the client.

Essentially, Kestrel is the **entry and exit point** of your application's HTTP processing — a lightning-fast bridge between the network and your managed code.

Why Kestrel Matters

Kestrel is:

- **Cross-platform** — runs on Windows, Linux, and macOS.
- **High-performance** — built on top of the asynchronous libuv and later Sockets.
- **Flexible** — can run standalone or behind a reverse proxy (like Nginx, IIS, or Apache).

When combined with the middleware pipeline, you get a powerful, customizable web stack that's both minimal and performant.

Final Thoughts

Kestrel and middleware together form the foundation of ASP.NET Core's web pipeline.

- Kestrel: Handles the network.
- Middleware: Shapes the request/response flow.

This architecture gives developers fine-grained control, speed, and scalability — making ASP.NET Core one of the most efficient and flexible web frameworks today.